

# Jatin

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## Summary

C++ Systems Engineer specializing in **multithreading, networking, and Windows internals**, with experience building **low-latency, high-throughput client systems**. Strong foundation in **concurrency, IPC, and distributed communication patterns**.

## Education

Dr B R Ambedkar National Institute of Technology

Aug 2019 - July 2023

Bachelor of Technology - Information Technology

## Technical Skills

**Programming Languages:** C++, C, Python, Java, C#

**Systems & Low-Level:** Win32 API, COM, Multithreading, IPC, OS Internals

**Backend & Networking:** REST APIs, WebSockets, TCP/IP, TLS/SSL, SQLite

**Frameworks & Libraries:** STL, GTest, POCO, PyTest

**Tools & Build Systems:** CMake, Vcpkg, MSVC, Git, Jenkins, Visual Studio, VSCode

**Debugging & Profiling:** WinDBG, ProcMon, Wireshark, SysInternal Tools

**Core Concepts:** Data Structures & Algorithms, Low-Level Design, System Design, Networking, Operating Systems

## Experience

Software Engineer, Zscaler

Aug 2024 – Present

- Designed a **high-performance C++ event delivery framework** enabling real-time security updates across clients
- Integrated **SQLite (C++)** as an **in-memory data layer** for notification workflows, enabling **low-latency local persistence**
- Distributed as a **vcpkg-managed library**, ensuring modularity, versioning, and independent integration
- Built **low-latency bidirectional communication** between native **Win32 host** and WebView UI using HostObjects
- Implemented **persistent WebSocket communication over TLS**, reducing polling overhead and lowering update latency
- Developed **non-blocking workflows using modern C# (async/await, Task-based model)**, eliminating UI thread blocking and improving responsiveness
- Revamped App UI using **WinUI3 + MVVM architecture**, designing **asynchronous view-model interactions** and scalable UI abstractions
- Improved system modularity and testability using **dependency injection and factory patterns**

Associate Software Engineer, Zscaler

Aug 2023 – Aug 2024

- Migrated **blocking RPC** to **asynchronous RPC**, reducing UI thread blocking and improving responsiveness under load
- Developed **multithreaded C++ components** using atomics and lock-free techniques, improving throughput in concurrent workflows
- Diagnosed and resolved production crashes via **WinDBG and dump analysis**, improving client stability across releases
- Automated regression workflows using **PyTest**, reducing manual testing effort by **25 %**
- Delivered optimized modules using **modern C++ (STL, RAII, concurrency primitives)**

Software Engineer Intern, Zscaler

Feb 2023 – Aug 2023

- Increased unit test coverage from **10% to 35%** using GTest for C++ codebase
- Built Windows UI components using **C#, WPF, XAML**
- Worked with **Windows APIs and asynchronous programming models** in C#

## Projects

Concurrent Task Scheduler | Modern C++, Multithreading, Atomics

[Code](#)

- Engineered a **thread-pool task scheduler with priority-based scheduling (4-level priority queue)**, processing **40K+ tasks/sec** with **3.75× speedup on 4 cores**
- Implemented **lock-based synchronization (mutexes, condition variables, atomics)** with **sub-microsecond submission latency (~0.41 µs/task)** and near-zero scheduling overhead
- Designed **graceful shutdown with deadlock-free teardown**, leveraging **RAII resource management** and polymorphic task execution

## Achievements & Awards

- **LeetCode Knight (Top 6%)** - Peak rating 1900+
- **Open Source Contributor** - Canonical Mir display server, worked on various modern cpp improvements, resolving bug fixes and adding unit tests
- **Zscaler Spot Award** - Recognized for design contribution and delivering high quality code for notification framework.